SHIKHAR PANDAV

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SUMMARY

Backend engineer, ∼2 years on distributed Python services (FastAPI, gRPC, RabbitMQ, Redis, GCP) across banking and e-commerce engagements. Depth in concurrency control, idempotent pipelines, and caching. **LeetCode Guardian** — peak 2250, top 1% globally, rank **207**/**21,482** in a Weekly Contest | **Codeforces Expert** — peak 1633.

EXPERIENCE

Infosys Pune, India
Specialist Programmer (Power Programmer) *July 2024 – April 2026*

E-commerce engagement — order-processing path

- Owned the order-processing path end to end, migrating it from synchronous in-process handoffs to an **at-least-once RabbitMQ** pipeline — idempotency-key dedup for safe retries, dead-letter queue for poison messages.
- Diagnosed burst-traffic queueing on that path to **async connection-pool saturation** via Prometheus histograms; right-sized pool limits and timeouts to clear head-of-line blocking without adding replicas.

Banking engagement — read path and service boundaries

- Owned the read path of a high-traffic transactional API, cutting database load with a **Redis read-through cache** — tenant-scoped keys and per-endpoint TTLs bounding staleness without write-path invalidation.
- Hardened its service boundaries by migrating inter-service REST/JSON to **gRPC**/**Protobuf**, moving contract breakage from runtime to build time via generated stubs and CI schema checks.

PROJECTS

FORGE — Self-Hosted LLM Gateway & Agentic CLI
Python, FastAPI, Redis, PostgreSQL | GitHub *June 2026 – Present*

- Built an **OpenAI-compatible API gateway** keeping the agent core provider-agnostic behind a single `LLMClient` protocol.
- Eliminated **read-compute-write races** in distributed token-bucket rate limiting with an atomic Redis Lua script.
- Layered Redis exact-match and **embedding-based semantic caching**, backed by per-model circuit breaking and backoff failover.
- Guarded tool execution with a 5-mode approval engine, path-escape checks, and sub-agent isolation; **55 tests** in GitHub Actions CI.

MyTube — Full-Stack Video Streaming Platform
React, TypeScript, Next.js, tRPC, PostgreSQL, Mux | GitHub *Nov 2024 – Jan 2025*

- Shipped a video platform with **Mux adaptive-bitrate** transcoding behind an end-to-end typed tRPC API over PostgreSQL.
- Fetched threaded comment trees in one round trip using **recursive CTEs**, eliminating N+1 queries.

SKILLS

Languages Python, C++, TypeScript, JavaScript, SQL

Backend FastAPI, Django, asyncio, SQLAlchemy, gRPC/Protobuf, RabbitMQ, Redis, PostgreSQL

Systems Microservices, Event-Driven Architecture, Caching, Rate Limiting, Idempotency, Concurrency Control, System Design (HLD/LLD)

Infra Docker, GCP, GitHub Actions CI, Prometheus, Grafana, pytest, Git, Linux

GenAI LLM Gateways, Agent Loops, Semantic Caching, Embeddings, Vector Databases, RAG

EDUCATION

KIET Group of Institutions, AKTU Delhi-NCR, India
B.Tech, Computer Science and Engineering *August 2020 – June 2024*